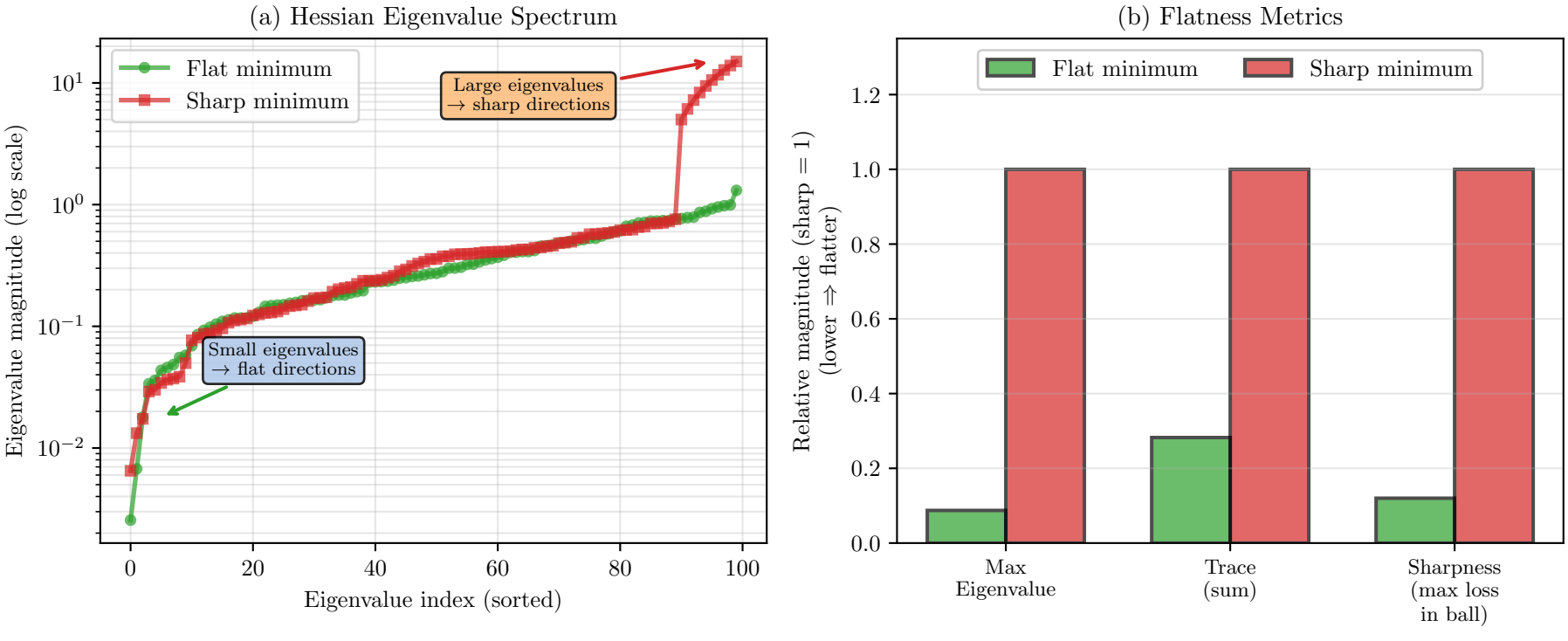
(a) Hessian Eigenvalue Spectrum
Flat minimum
Sharp minimum
Large eigenvalues
→ sharp directions
Small eigenvalues
→ flat directions
Eigenvalue magnitude (log scale)
Eigenvalue index (sorted)
(b) Flatness Metrics
Flat minimum
Sharp minimum
Relative magnitude (sharp = 1)
(lower ⇒ flatter)
Max
Eigenvalue
Trace
(sum)
Sharpness
(max loss
in ball)